

Project Thesis

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I. INTRODUCTION

THE aim of the project is the sizing of a power conversion system for xEV.

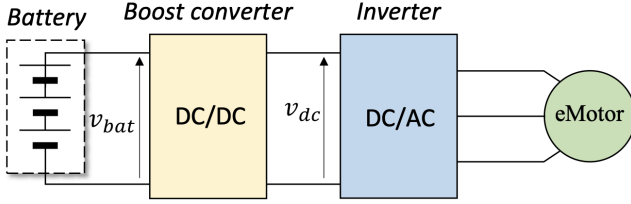


Fig. 1. xEV power conversion system

The considered power conversion system consists of:

- Traction HV battery
- DC-DC boost converter (single bi-directional switching cell)
- Three-phase inverter
- Synchronous electrical motor

The project will follow this taxonomic procedure:

- 1) Sizing of the input inductor of the DC-DC converter. Computation of the voltage stress for power devices and define the rated voltage of power devices. Computation and definition of the current stress for the power devices and the input inductor
- 2) Design of the DC-DC converter cascaded control.
- 3) Simulation verification DC-DC converter operation.
- 4) Retrieving the machine type and fundamental parameters based on the given data.
- 5) Sizing the inverter based on the motor data and operating cycle.
- 6) Calibrating a FOC algorithm for the provided motor.
- 7) Implementing in Simulink the model of the machine and testing the performances of the drive in the operating cycle without the DC-DC converter.
- 8) Simulation of the whole system in Simulink for at least one machine operating point, using average dynamic models for DC-DC converter and inverter.

The given data is contained in the file *Data11.mat*:

- *Cycle_n*: mechanical speed in rpm;
- *Cycle_T*: related torque request;
- *DCDC_Co*: output capacitance of the DCDC converter (μF);
- *DCDC_dIbatt*: input current ripple (in percentage) with respect to the mean input rated current;
- *DCDC_fsw*: switching frequency (kHz);

- *DCDC_Pmax_kW*: rated power (kW);
- *DCDC_vbatt*: rated output voltage (V);
- *DCDC_vout*: battery voltage (V);
- *Fd*: matrix that provides λ_d ;
- *Fq*: matrix that provides λ_q ;
- *Id* and *Iq*: regularly spaced grid of points in the plane;
- *id_KtMax*, *id_KvMax*, *iq_KtMax*, *iq_KvMax*: These vectors are the currents needed to find the MTPA and MTPV trajectories;
- *p*: number of pole pairs;
- *Rs*: stator resistance;

II. DC-DC CONVERTER

A. Input inductor

In order to compute the inductance value of the DC-DC converter, some preliminary quantities must be calculated:

Duty cycle

$$d = \frac{DCDC_vbatt}{DCDC_vout} = 0.5 \quad (1)$$

Regarding the duty cycle computations, it has been followed the upper transistor convention, similarly to what occurs when analyzing a buck converter.

Switching frequency

$$f_{sw} = DCDC_f_{sw} * 1000 = 18000 \quad [Hz] \quad (2)$$

Output power

$$P_{out} = DCDC_Pmax_kW * 1000 = 85000 \quad [W] \quad (3)$$

Input power

$$P_{in} = \frac{P_{out}}{\eta} = 89474 \quad [W] \quad (4)$$

P_{in} and P_{out} are connected together by the efficiency parameter η which has been estimated to be $\eta = 0.95$.

Input rated current

$$I_{in_rated} = \frac{P_{in}}{DCDC_vbatt} = 447.37 \quad [A] \quad (5)$$

The input rated current corresponds to the average inductor current.

Current ripple

$$\Delta I_{in} = \frac{I_{in_rated} \times DCDC_dIbatt}{100} = 35.79 \quad [A] \quad (6)$$

Inductance

$$L = \frac{1}{\Delta I_{in}} \times \frac{DCDC_vout \times (1 - d) \times d}{f_{sw}} = 1.552 \times 10^{-4} \quad [H] \quad (7)$$

B. Current and voltage stress

Average current on the inductor

$$I_{L,avg} = I_{in,rated} = 447.37 \quad [A] \quad (8)$$

RMS current ripple on the inductor

$$\Delta I_{L,rms} = \frac{\Delta I_{in}}{2 \times \sqrt{3}} = 10.33 \quad [A] \quad (9)$$

Average current on the inductor

$$I_{L,rms} = \sqrt{I_{L,avg}^2 + \Delta I_{L,rms}^2} = 447.49 \quad [A] \quad (10)$$

Load current

$$I_{load} = \frac{P_{out}}{DCDC_{vout}} = 212.5 \quad [A] \quad (11)$$

When $i_L < 0$ and $q = 1$, D1 is conducting, while when $i_L < 0$ and $q = 0$, T2 is conducting.

Since the request is to compute the current stress for a boost converter we are always in these 2 cases.

Average current on D1

$$I_{D1,avg} = d \times I_{L,avg} = 223.68 \quad [A] \quad (12)$$

RMS current on D1

$$I_{D1,rms} = \sqrt{d} \times I_{L,rms} = 316.42 \quad [A] \quad (13)$$

Average current on T2

$$I_{T2,avg} = (1 - d) \times I_{L,avg} = 223.68 \quad [A] \quad (14)$$

RMS current on D1

$$I_{T2,rms} = \sqrt{1 - d} \times I_{L,rms} = 316.42 \quad [A] \quad (15)$$

Average current on capacitor

$$I_{C,avg} = 0 \quad [A] \quad (16)$$

The average current on capacitor must be null in order to attain capacitor charge balance.

RMS current on capacitor

$$I_{C,rms} = \sqrt{I_{D1,rms}^2 + I_{load}^2} = 234.45 \quad [A] \quad (17)$$

Voltage on power devices

$$V_{power,devices} = DCDC_{vout} = 400 \quad [V] \quad (18)$$

According with this kind of voltage stress and switching frequency we can adopt both IGBTs (slower but higher maximum voltage) or MOSFET (faster but lower maximum voltage) as transistors.

C. Cascaded control

Regarding the design of the inner current control loop, it has been adopted an open loop approach.

First of all, the crossover frequency has been equated to 1/10 of the switching frequency in order to ensure an decent phase margin, then the proportional gain might be imposed:

Crossover frequency

$$\omega_{c,i} = \frac{2\pi \times f_{sw}}{10} = 11310 \quad [rad/s] \quad (19)$$

Proportional gain

$$k_{p,i} = L \times \omega_{c,i} = 1.7556 \quad (20)$$

$\omega_{c,i}$ is approximated as $k_{p,i}/L$ since the converter pole occurs at higher frequency.

The following step is meant to impose the controller zero at 1/10 of the converter pole.

Delay introduced by the converter

$$\tau_d = \frac{1.5}{f_{sw}} = 8.333 \times 10^{-5} \quad [s] \quad (21)$$

Zero frequency

$$\omega_{z,i} = \frac{1}{10 \times \tau_d} = 1200 \quad [rad/s] \quad (22)$$

Integral gain

$$k_{i,i} = \omega_{z,i} \times k_{p,i} = 2106.7 \quad (23)$$

It is remarkable to check the crossover frequency and the phase margin by means of the Bode plot of the correspondent transfer function of the system to be stabilized:

Inner loop control transfer function

$$HOL_i = \frac{k_{p,i} \times (s + \omega_{z,i})}{L \times s^2 \times (1 + s \times \tau_d)} \quad (24)$$

The Bode plot provides the following results:

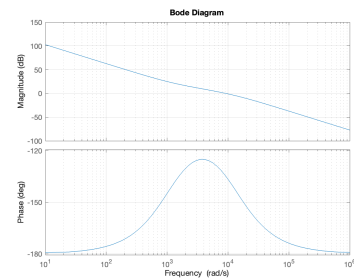


Fig. 2. Inner loop Bode plot

Regarding the design of the outer voltage control loop, it has been adopted an open loop approach.

TABLE I. Inner loop Bode plot results

Symbol	Quantity	Unit
$\omega_{b,i}$	11500	rad/s
$\omega_{c,i}$	9100	rad/s
Φ	45.3	deg

First of all, the crossover frequency has been equated to 1/10 of the current, which is 1/10 of the switching frequency in order to obtain an outer loop slower than the inner loop:

Crossover frequency

$$\omega_{c,v} = \frac{2\pi \times f_{sw}}{100} = 1131 \quad [rad/s] \quad (25)$$

Proportional gain

$$k_{p,v} = DCDC_C \times 10^{-6} \times \omega_{c,v} = 0.6786 \quad (26)$$

Then it has been imposed a 60 degrees phase margin imposing $\omega_{z,v}$:

Zero frequency

$$\omega_{z,v} = \frac{\omega_{c,v}}{sqrt{3}} = 652.97 \quad [rad/s] \quad (27)$$

Integral gain

$$k_{i,v} = \omega_{z,v} \times k_{p,v} = 443.09 \quad (28)$$

It is remarkable to check the crossover frequency and the phase margin by means of the Bode plot of the correspondent transfer function of the system to be stabilized:

Outer loop control transfer function

$$HOL_v = \frac{k_{p,v} \times (s + \omega_{z,v})}{DCDC_C \times 10^{-6} \times s^2} \quad (29)$$

The Bode plot provides the following results:

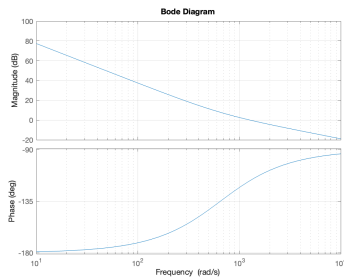


Fig. 3. Outer loop Bode plot

TABLE II. Outer loop Bode plot results

Symbol	Quantity	Unit
$\omega_{b,v}$	1700	rad/s
$\omega_{c,v}$	1270	rad/s
Φ	63	deg

Simulation and results

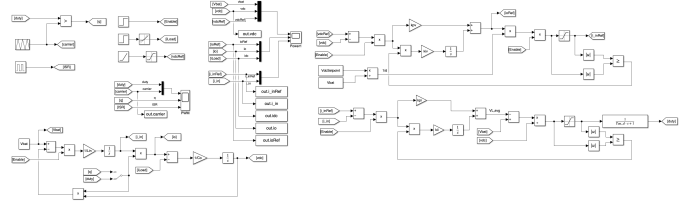


Fig. 4. Simulink setting for DC/DC converter control

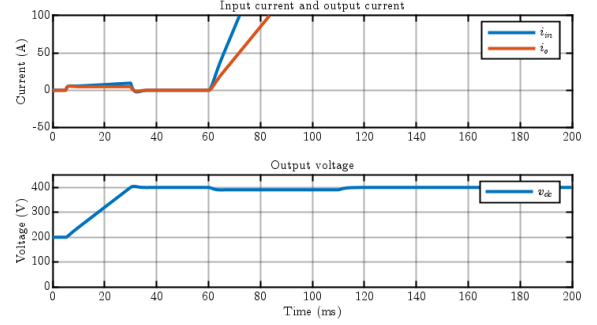


Fig. 5. Simulation results. From top to bottom: input and load currents (A), DC voltage (V)

When **enable** is not active nothing moves, both the loops are blocked and $i_{out,Ref}$, $i_{in,Ref}$ are null, V_{dc} and i_{in} are defined by their initial conditions, V_{bat} is constant.

Since $V_{dc} = V_{bat}$, the duty cycle rapidly saturates to 1 at the beginning of the time course.

The larger bandwidth of the first controller allows it to work faster, as predictable in the way in which $i_{in,Ref}$ is better tracked by i_{in} with respect to $V_{dc,Ref}$ and V_{dc} .

Considering the overall switching dynamics, when $q = 1$, the output current equates the input current, and $V_A = V_{dc}$, determining a negative voltage on the inductor $V_L < 0$ and input and output currents to decrease; while for $q = 0$ output current is null, $V_A = 0$ and the voltage on the inductor equates the battery voltage ($V_L = V_{bat}$) causing the input current to increase.

As a general rule, the input current can always be seen as the sum of the mean value determined by the load current or dc current (or even by the capacitor in the transitory phases) and a ripple component due to the switching dynamic and in particular to the voltage applied to the inductor during the switching operation.

When **enable** and **VdcSetPoint** are activated, the voltage loop ask for more output current in order to charge the capacitor, causing a subsequent increase of the input current request, which in turn increases the error between the reference and the actual input current. The average voltage on the inductor $V_{L,avg}$ increases while $V_{A,avg}$ decrease, the duty cycle decreases and reaches the steady state value at 0.5.

When **VdcSetPoint** is reached at $t = 30ms$ the capacitor is charged and since no load current is present, the capacitor current is equal to the output current and, according to the charge balance, the output current mean value is zero. Since

for the average dynamical model $i_o = d * i_{in}$, even $i_{in,avg}$ will be null at steady state in this very same conditions. At steady state the duty cycle is 0.5 because we must keep $V_o = V_{in}/d$. The current peaks are determined by the current ripple and since the mean value is zero, they are equal to $\pm \Delta I_{in}/2$, which determines a small voltage ripple.

When at $t = 60ms$ the load current is added, the capacitor start discharging and the voltage V_{dc} decreases. The voltage loop asks for more output current in order to maintain the capacitor charge, the reference input current increases and the average voltage on the inductor $V_{L,avg}$ increases as well. The duty cycle mean value, however, is maintained at 0.5 to ensure $V_o = V_{in}/d$. Therefore the input current ripple is maintained as before.

When at $t = 110ms$ the load current reaches its maximum value, the input current oscillates instantaneously around the mean rated value of 425A (considering the converter model ideal) with the same ripple determined by the correspondent duty cycle.

III. MOTOR MODEL

This section must include motor rated parameters (as input data), key equations and description of direct and inverse flux maps, as well as MTPA profiles.

A. Motor data

The motor parameters are provided in the following table, in order we have: the nominal inductances (apparent and differential) along the synchronous axes (d,q), the magnetic flux of the permanent magnets.

TABLE III. Motor data.

Symbol	Quantity	Unit
L_d-nom	$1.21 \cdot 10^{-4}$	H
L_q-nom	$4.94 \cdot 10^{-4}$	H
l_d-nom	$1.10 \cdot 10^{-4}$	H
l_q-nom	$3.45 \cdot 10^{-4}$	H
λ_m	0.0573	Wb

B. Motor equations

The voltage equations in dq coordinates for a synchronous machine is

$$v_{dq} = R_s i_{dq} + \frac{d}{dt} \lambda_{dq} + j\omega \lambda_{dq} \quad (30)$$

And the magnetic model of an IPM machine is

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \cdot \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \lambda_m \\ 0 \end{bmatrix} \quad (31)$$

We also computed the apparent inductances as follows:

$$L_d = \frac{\lambda_d(i_d, i_q = 0)}{i_d} \quad (32)$$

$$L_q = \frac{\lambda_q(i_d = 0, i_q)}{i_q} \quad (33)$$

and also the differential inductances:

$$l_d = \frac{\partial \lambda_d(i_d, i_q)}{\partial i_d} \quad (34)$$

$$l_q = \frac{\partial \lambda_q(i_d, i_q)}{\partial i_q} \quad (35)$$

In order to evaluate inductances, according to the magnetic model, we had to remove from the total flux the one comes from the magnet. What we got are the differential inductances superimposed with the apparent ones in the first quasi-linear line. And also we obtained, as expected, L_d much lower than L_q . We selected nominal values (a good trade-off between linear trait and saturation region) as follows:

$$L_d = 1.21 \cdot 10^{-4} \text{ H}, L_q = 4.94 \cdot 10^{-4} \text{ H}, l_d = 1.10 \cdot 10^{-4} \text{ H}, l_q = 3.45 \cdot 10^{-4} \text{ H} \quad (36)$$

We report here, in order to conclude, the plots of the apparent and differential inductances on varying the current. Aware of the symmetry with respect to the current axis we did our considerations only for positive currents.

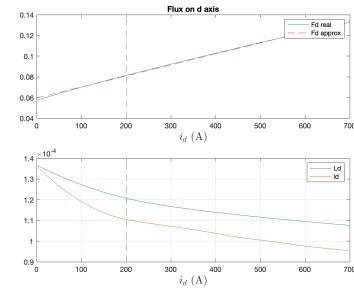


Fig. 6. Plot of the flux, apparent and differential inductances on d-axis

As we can see, in this above figure is reported the plot of the Flux on d-axis with its approximation; furthermore, it is also reported the trend of the inductances varying the current. We have also done the same thing on the q-axis reporting the two subplots for the flux and inductances.

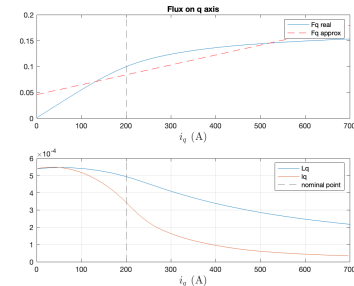
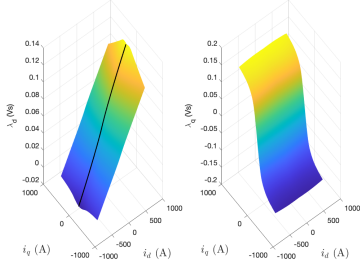


Fig. 7. Plot of the flux, apparent and differential inductances on q-axis

C. Flux maps and MTPA

This subsection is dedicated to direct and inverse flux maps and also for MTPA profiles. The Flux maps (obtained from the set of matrices included in the given data) are shown in the figures below. They represent the dependence of λ_d and λ_q on

Fig. 8. Flux maps: $\lambda_d(i_d, i_q)$ and $\lambda_q(i_d, i_q)$

i_d and i_q . The iron saturation level is affected by the current on both axes and this effect is called “cross saturation”.

Looking at the current matrices, we noticed that the row 256 of i_q is null and the column 256 for i_d is null. Plotting the relative flux entries in those positions, we evaluated the flux obtained with the contribute of only one of the two currents. For zero currents, we obtain a flux on the “d” direction. This means that we will have an SPM or IPM, because for PM-SyR we would have obtained a negative flux in the “q” axis, while for a SyR we would not have found any flux (both in d and q axis due to the lack of PMs).

Now, we report the MTPA and MTPV plots. The MPTA is the trajectory in the dq plane corresponding to the maximum torque for a given amplitude of the current vector.

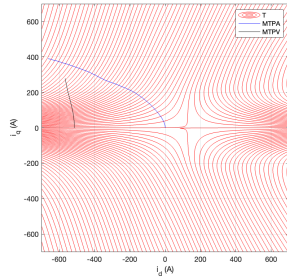


Fig. 9. MTPA and MTPV trajectories, Torque

It is also shown the MPTV trajectory that is the locus of the dq plane with the highest torque for a given flux amplitude. These trajectories are helpful to understand the type of motor. Since the MTPA and the MTPV are in the second quadrant we conclude we are in front of an IPM machine.

IV. INVERTER SIZING

The inverter current rating, the inverter DC-link voltage and the minimum switching frequency are the quantities that need to be found for the sizing of the inverter. The maximum current has been computed in order to guarantee a 10% safety margin, starting from the operating points as suggested in the project instructions. In, particular among the points of the driving cycle (figure 10) the one that require the most torque is (1645.58[rpm], 112.703[Nm]). Considering a 10% more we got a value of about 124Nm. We then followed the MTPA trajectory until it intercepted the isotorque curve of 124Nm.

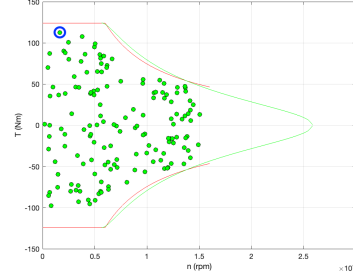
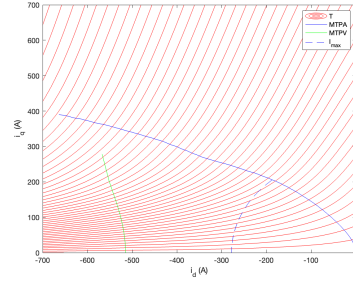


Fig. 10. Driving cycle with the maximum torque point highlighted

This point corresponds to a particular current vector whose module gives the inverter’s rated current.

$$I_{max} = 277.8006A \quad (37)$$

Fig. 11. MTPA, MTPV, Torque and I_{max} characteristics

The DC-link voltage has been found through the relation

$$V_{max} = |\lambda_{dq}| \cdot \omega_{base} \quad (38)$$

The flux linkage has been found interpolating the flux maps with the rated current (d, q) components. The base speed has been imposed of 1885 rad/s (6000rpm). This choice has been taken in order to limit the output voltage of the DCDC converter and at the same time to guarantee a good fitting of driving cycle by the inverter operating limits. The obtained V_{dc} and V_{max} are the following:

$$V_{max} = 212.4V \quad V_{dc} = \sqrt{3} \cdot V_{max} = 367.9V \quad (39)$$

Finally, the minimum switching frequency has been computed considering the maximum speed of the driving cycle contained in the vector $Cycle_n$:

$$f_{s,min} = 22 \cdot \frac{rpm_{max}}{60} \cdot p = 16.57kHz \quad (40)$$

V. DESIGN OF FOC CONTROLLERS WITH FEED-FORWARD OF THE MOTIONAL TERM

The design of the FOC controller is straightforward. The starting point are the LUT tables from which based on the required torque, the reference current in the (d,q) frame are retrieved. Then, a couple of PI regulators are implemented in order to ensure the stability provided by the feedback of the measured current. The parameters of such regulators are designed according to the open-loop approach. Therefore the bandwidth/crossover frequency of is set at one tenth of

the sampling frequency. Afterwards the proportional gains are computed exploiting the approximation

$$\omega_b = \frac{k_{pj}}{l_j} \quad j = d, q \quad (41)$$

Finally the integral gains are set just imposing:

$$k_{ij} \ll k_{pj} \cdot \omega_b \quad (42)$$

An additional request was to implement a feed-forward block of the motional term. To do that, it is sufficient to start from the current in the (d,q) frame, exploit the magnetic model and retrieve the flux in the (d,q) frame and finally to multiply for the electrical speed. These terms are then added at the end of the PI regulators block.

Simulations and results

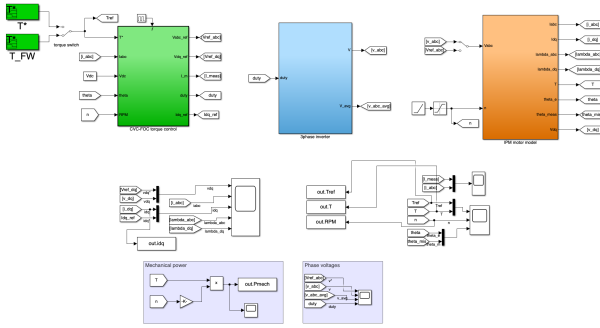


Fig. 12. Simulink setting for IPM machine control

A. Below base speed

The objective of the first simulation is to analyze the behaviour of the IPM machine working together with the inverter, when the load conditions vary from no load up to maximum torque.

The first step requires to set the manual torque switch to the upper switch in order to have a ramp torque reference varying from a null value to the maximum torque required in the operating cycle.

Then, it is necessary to set the manual current switch to the upper switch in order to match the current reference extracted from the LUT.

The motor works as a torque generator so the speed is imposed by the load. In this configuration, the speed varies within a range of $[0, -\omega_b]$ with a slope that allows to reach the base speed within the simulation time.

Torque and reference torque

The torque T adequately follows the reference torque T_{ref} : both follow a linear trend until $t = 300ms$, then the torque plateau is reached at $T = 112.5$ Nm.

Mechanical speed and base speed

The base speed represents a constant value, seen as ultimate limit for what concerns the mechanical speed.

The latter follows a linear trend between $t_0 = 100ms$ and $t = 400ms$, starting after a certain delay and reaching the

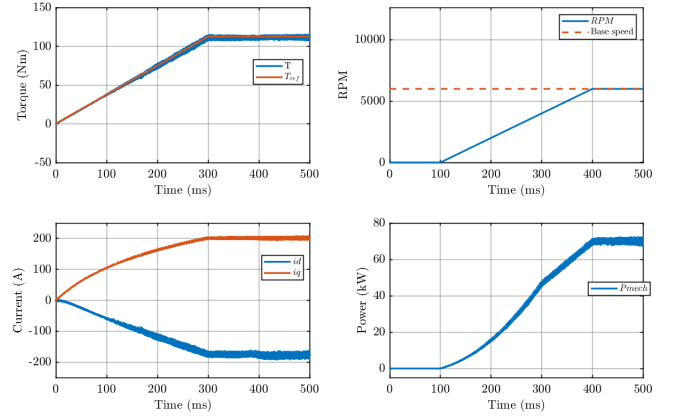


Fig. 13. Simulation results: T and T_{ref} (top left), RPM and base speed (top right), i_d and i_q (bottom left), P_{mech} (bottom right).

steady state at the value of 6000 rpm (ω_b).

d-frame and q-frame currents

As predictable, i_d and i_q currents follow two almost specular trends: while i_d reaches the steady state at $t = 300ms$, i_q shows a negative trend to $t = 300ms$, right before reaching the steady state.

Mechanical power

P_{mech} starts growing at $t_0 = 100ms$ and reaches steady state conditions at $t = 400ms$, with power values around 70 kW.

B. Field weakening

The objective of the second simulation is to analyze the behaviour of the IPM machine working in field weakening.

For this purpose, we have chosen a point in the driving cycle with coordinates: $[11483RPM, 40Nm]$.

The first step requires to set the manual torque switch to the lower switch in order to have a constant torque reference of magnitude $T = 40$ Nm.

Then, it is necessary to set the manual current switch to the lower switch in order to match the current reference extracted from the torque contours and the flux maps.

In this configuration, the speed is set to a ramp within a range of $[0 - 11483]$ RPM with a slope that allows to reach the desired speed within the simulation time.

TABLE IV. Field weakening point

Torque	Angular speed
40 Nm	11483 rpm

Torque and reference torque

The torque T adequately follows the reference torque T_{ref} : both are constant at an average value of $T = 40$ Nm, reached immediately after a step variation in $t = 0ms$.

Mechanical speed and base speed

The base speed represents a constant value, while the

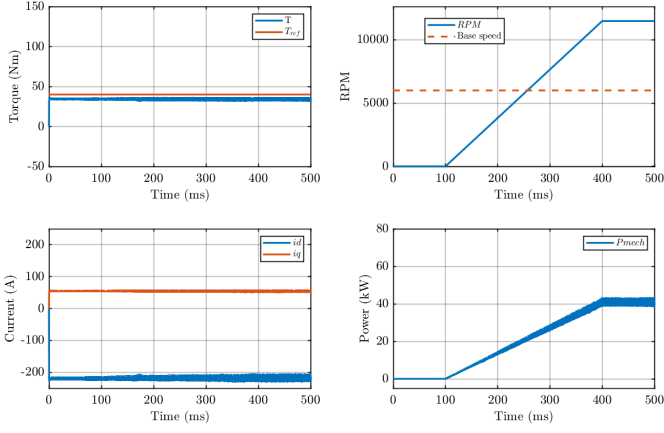


Fig. 14. Simulation results: T and T_{ref} (top left), RPM and base speed (top right), i_d and i_q (bottom left), P_{mech} (bottom right).

mechanical speed follows a linear trend between $t_0 = 100ms$ and $t = 400ms$, starting after a certain delay and reaching the steady state at 11483 rpm.

d-frame and q-frame currents

i_d and i_q currents are both constant, reaching the respective values of $i_d = 50$ A and $i_q = -210$ A, after a step variation in both cases.

Mechanical power

P_{mech} starts growing at $t_0 = 100ms$ and reaches steady state conditions at $t = 400ms$, with power values around 40 kW.

VI. SIMULATION RESULTS

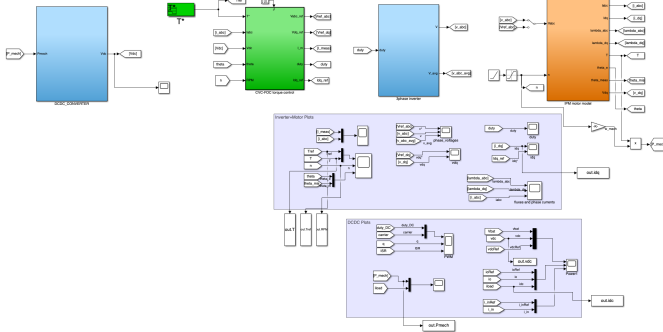


Fig. 15. Simulink settings of the integrated model

In order to merge the DC/DC converter, the motor and the inverter models, firstly, the $V_{dc_setpoint}$ in the DCDC converter subsystem has been set equal to the one associated with the required inverter DC link voltage previously found imposing the rated flux (associated with the maximum torque required) and the base speed.

Secondly, the I_{load}/I_{dc} should be modified adequately to constantly match the current requested by the inverter.

The key factor is to start from the output mechanical power, obtained simply as the product of the motor torque with the mechanical speed; then, assuming an efficiency of 90% for

what concerns the condensed block inverter + IPM machine, the output power necessary to the DC/DC converter can be determined.

Ultimately, such power is divided for the V_{dc} voltage, in order to obtain the I_{load}/I_{dc} , which subtracted to the output current gives the input capacitor current. Integrating such value and imposing an initial condition for the capacitor voltage, the output voltage V_{dc} can be retrieved.

It has been chosen to select a working point which is right one point below the base speed; in this way, it is still possible to work in MTPA conditions:

TABLE V. Working point

Torque	Angular speed
100.6 Nm	2459 rpm

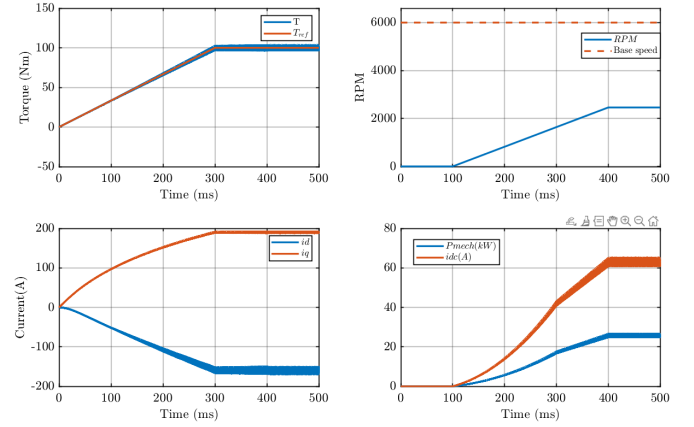


Fig. 16. Simulation results: T and T_{ref} (top left), RPM and base speed (top right), i_d and i_q (bottom left), P_{mech} and i_{dc} (bottom right).

Torque and reference torque

The torque T adequately follows the reference torque T_{ref} : both follow a linear trend until $t = 300ms$, then the torque plateau is reached at $T = 100$ Nm.

Mechanical speed and base speed

The base speed represents a constant value, seen as ultimate limit for what concerns the mechanical speed.

The latter follows a linear trend between $t_0 = 100ms$ and $t = 400ms$, starting after a certain delay and reaching the steady state at the value of 2459 rpm.

d-frame and q-frame currents

As predictable, i_d and i_q currents follow two almost specular trends: while i_d reaches the steady state at $t = 300ms$ with a quasi-linear attitude, i_q shows a negative linear trend to $t = 300ms$, right before reaching the steady state.

Mechanical power and DC current

Both P_{mech} and i_{dc} start growing at $t_0 = 100ms$ and reach steady state conditions at $t = 400ms$, with power values around 25 kW and current on the order of 60 A.

VII. CONCLUSIONS

In this project, we have meticulously designed and analyzed a power conversion system for electric vehicles (xEVs), integrating a high-voltage traction battery, a DC-DC boost converter, a three-phase inverter, and a synchronous electrical motor. The process followed a structured taxonomic procedure encompassing several critical stages from the sizing of the DC-DC converter to the simulation of the entire system. The project successfully achieved its objective of designing a robust and efficient power conversion system for xEVs. The complete analysis and simulation results confirm the effectiveness of the proposed system components and control strategies.